GEOMETRY · AREA, SURFACE AREA AND VOLUME

Solids Volume Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Solids Volume Lab** on Mr. K's Classroom: Geometry → Area, Surface Area and Volume → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. A cone and a cylinder have the same radius and height. How many cones would fill the cylinder?

   3

2 Explore use the tool and record what you see

Volume and surface area

Leave π in your answers where it appears.

Solid	Dimensions	Volume	Surface area
Rectangular prism	ℓ = 5, w = 3, h = 4	**60**	**94**
Cylinder	r = 3, h = 6	**54π ≈ 169.6**	**54π ≈ 169.6**
Cylinder	r = 3, h = 8	**72π**	**66π**
Cone	r = 3, h = 6	**18π ≈ 56.5**	**≈ 91.5**
Square pyramid	s = 6, h = 4	**48**	**96**
Sphere	r = 3	**36π ≈ 113.1**	**36π ≈ 113.1**

Scale the 5 × 3 × 4 prism

k	Volume	Volume ÷ 60	Surface area	SA ÷ 94
1	60	1	94	1
2	**480**	**8**	**376**	**4**
3	**1620**	**27**	**846**	**9**

3 Write the rule fill in each blank from what you observed

1. For a prism or cylinder, V = **Bh (base area × height)**.
2. For a pyramid or cone, V = **$\frac{1}{3}Bh$**.
3. A cone holds **$\frac{1}{3}$** of a cylinder with the same radius and height.
4. For a sphere, V = **$\frac{4}{3}\pi r^3$** and SA = **$4\pi r^2$**.
5. Scaling every dimension by k multiplies surface area by **k^2** and volume by **k^3**.

4 Check no tool: show your work

1. Find the volume of a cylinder with r = 5 and h = 10.

   250π

2. Find the volume of a cone with r = 6 and h = 4.

   48π

3. Find the volume of a sphere with diameter 12.

   288π

4. Two similar solids have a scale factor of 3. Compare their volumes.

   The larger is 27 times the smaller

5 Reflect

Why does doubling every dimension multiply the volume by 8?

Volume uses three dimensions, each doubled: 2 · 2 · 2 = 8.